

Chapter 35

Ultrasound Contrast Agents

Mark A. Borden^{1,2}, Paul A. Dayton³, Connor Slagle¹ and Rachel W. Walmer³

¹Department of Mechanical Engineering, University of Colorado, Boulder, CO, United States; ²Biomedical Engineering Program, University of Colorado, Boulder, CO, United States; ³Joint Department of Biomedical Engineering, University of North Carolina and North Carolina State University, Chapel Hill, NC, United States

Introduction

Due to its safety, portability, and low cost, ultrasound imaging is widely used for clinical applications spanning cardiology, peripheral vascular disease, obstetrics, kidney diseases, and cancer. Ultrasound molecular imaging (USMI) is becoming increasingly popular for small animal studies [1,2], and the first human trials were reported in 2017 for diagnosis of prostate and breast cancer [3,4]. Ultrasonic blood echoes are two to three orders of magnitude smaller than tissue echoes due to the relatively small acoustic impedance difference between red blood cells and plasma. In order to detect small blood vessels and receptors within these vessels, ultrasonic contrast agents have been developed. Several types of particles with varying shell and core properties have been engineered, although fundamentally all ultrasound contrast agents can be simplified to the presence of a gas core, which provides an acoustic signal. The degree of acoustic backscatter depends on the intrinsic properties of the ultrasound contrast agent particle. The compressibility of the particle and the density difference between the particle and surrounding tissue contribute

to the acoustic backscatter. Nonlinear effects and resonance can also contribute considerably to the echo response and contrast-to-tissue ratio.

One advantage of the ultrasound contrast agent is its acoustic response, making it amenable to the dual purposes of imaging and therapy with external ultrasound transducers [5]. Another benefit is the surface area for targeting and payload. Ultrasound contrast agents range in size between 0.1 and 10 μm diameter and therefore can be engineered to express many ligands. Multiple ligand-receptor interactions can lead to firm adhesion, even in the face of hydrodynamic forces acting to dislodge the contrast agent. The ultrasound contrast agent can also carry a payload of other contrast agents for multimodality (fluorescence [6], magnetic resonance [7], or photoacoustic [8]) imaging, or therapeutic molecules for theranostic applications [9,10].

This chapter will introduce the contrast agent particles that are being used as molecular imaging probes for ultrasound, namely microbubbles and phase-change nanodrops, with the structures summarized in Fig. 35.1. Emphasis is placed on the most commonly used echocontrast agent, the

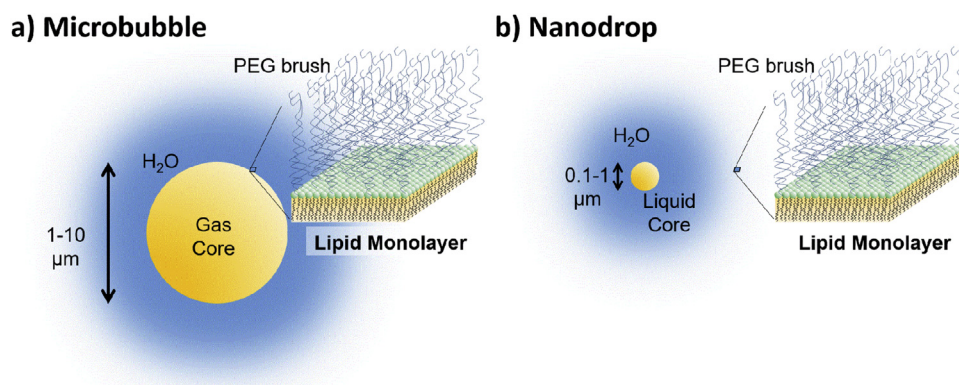


FIGURE 35.1 Ultrasound contrast agents.

microbubble. Almost all USMI studies reported so far in the literature have employed microbubbles. The theory of microbubble acoustics is now highly developed and experimentally validated [11–14], revealing a rich library of useful physical phenomena. The key equations relevant to USMI are provided in this chapter. Microbubble synthesis and ligand conjugation strategies are also discussed. Interactions between the microbubble and ultrasound device are introduced in the context of methods used to exploit them for the purposes of molecular imaging. Details of ultrasound instrumentation, which is used to excite and detect microbubbles in vivo, are covered in Chapter 11, “Ultrasound.” We also consider examples of preclinical and clinical applications for USMI, as well as safety considerations. We end with a statement on current limitations and challenges, as well as opportunities and future directions.

The Echogenic Microbubble

Microbubble Acoustics

The microbubble (MB) is an ideal ultrasound contrast agent because it is extremely echogenic, as well as being biocompatible, multifunctional, and economical [5]. MBs are the only ultrasound contrast agents to have been approved by the U.S. Food and Drug Administration (Table 35.1); the first approval came for Albunex in 1997. MBs are gas spheres typically 1–6 μm in diameter, with the lower limit determined by echogenicity and the upper limit by transcapillary transport. MBs are much smaller than the wavelength of diagnostic ultrasound, which is typically 100–1000 μm , so they experience a nearly isotropic pressure field that changes with time. The gas core has a low density and is highly compressible, allowing it to shrink and expand with the passage of an acoustic wave. It is these radial oscillations that provide such excellent echogenicity [11].

An ordinary differential equation has been derived through the combined works of Rayleigh, Plesset, and others for the radial oscillations of a MB driven by a harmonic ultrasound pressure pulse. The dynamic model was

formulated by taking a force balance at the bubble wall to arrive at the following expression for an acoustically driven MB radius as a function of time [12]:

$$\ddot{R}R + \frac{3}{2}\dot{R}^2 + \frac{p_\infty - p_L}{\rho} = 0 \quad (35.1)$$

where R is the MB radius, the overdots represent first and second time derivatives, p_∞ is the background pressure and includes the hydrostatic pressure and acoustic driving pressure, p_L is the pressure at the liquid side of the bubble wall, and ρ is the liquid density. The nonlinear radius terms in Eq. (35.1) arise from the spherical geometry. Additional nonlinearity is introduced into the pressure terms by the polytropic gas model, viscosity in the surrounding liquid media, and viscoelastic terms for the encapsulating shell [12]. These terms are excluded here for the sake of brevity, but the interested reader may find more information in the following references [11–14], to name just a few.

An example calculation of the Rayleigh-Plesset equation for an uncoated MB driven by a typical ultrasound imaging pulse is shown in Fig. 35.2A and B. Increasing the pressure amplitude from 100 to 500 kPa results in a transition from a more linear response to a highly nonlinear response. With a higher ultrasonic driving pressure, the rate of collapse is very rapid, faster than the rate of expansion, owing to the liquid inertia. The MB radius increases and decreases at a rapid velocity giving rise to a strong and unique echo. The amplitude of the echo pressure p at a distance of r from the MB center is given by Ref. [12]:

$$p(r) = \frac{\rho R}{r} (\ddot{R}R + 2\dot{R}^2). \quad (35.2)$$

The backscattered pressure is again a nonlinear function of the MB radius, adding more nonlinearity to the overall acoustic response. The bubble echo can be predicted by solving the Rayleigh-Plesset Eq. (35.1) and using the result to calculate the pressure with Eq. (35.2). Fig. 35.2C shows how the MB acoustic response increases nonlinearly with driving pressure.

Fortuitously, the natural reaction time of a MB to a rapid pressure variation is on the order of nanoseconds to microseconds. Thus, MBs resonate at frequencies typically

TABLE 35.1 Reported doses for commercially available MB contrast agents.

Formulation	Shell/gas	Concentration (1/mL)	Mean diameter (μm)	Recommended dose ($\mu\text{L/kg}$) ^a
Optison (GE Healthcare)	Protein/C ₃ F ₈	5–8 $\times 10^8$	3.0–4.5	6
Definity (Lantheus)	Lipid/C ₃ F ₈	1.2 $\times 10^{10}$	1.1–3.3	10
Lumason (Bracco)	Lipid/SF ₆	1.5–5.6 $\times 10^8$	1.5–2.5	25

^aBolus intravenous injection into peripheral vein. Note that commercially available ultrasound contrast agents are polydisperse in size, thereby making ambiguous the classification by average diameter.

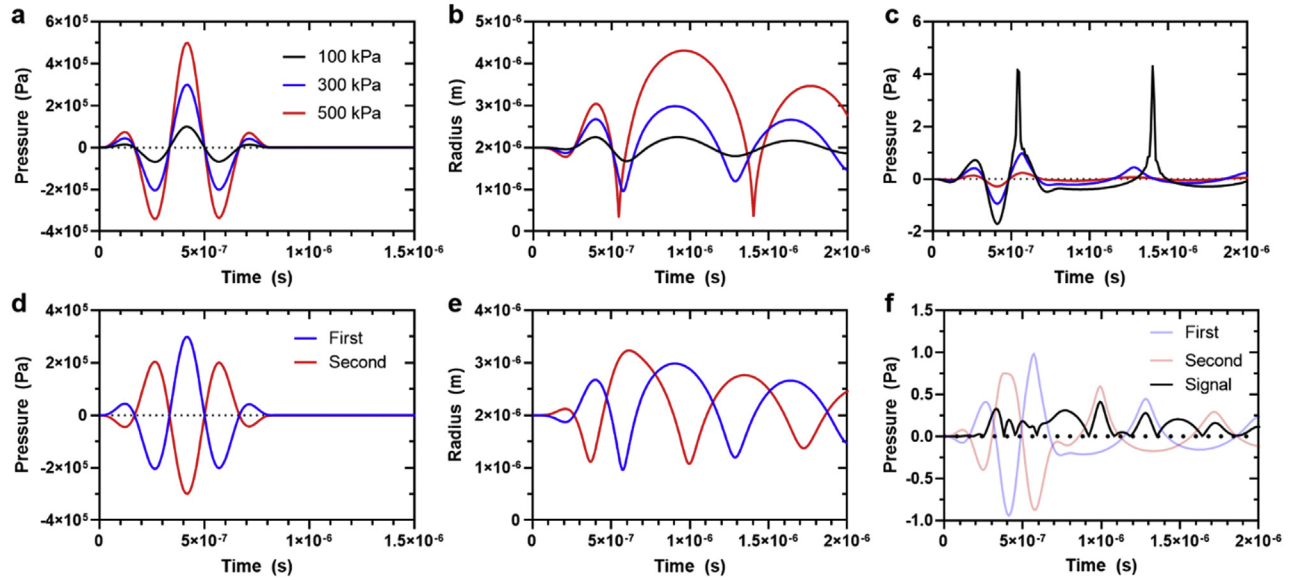


FIGURE 35.2 Microbubbles (MBs) can launch a nonlinear echo in response to an ultrasound imaging pulse. MB radial oscillations can be simulated with the Rayleigh-Plesset equation (Eq. 35.1) and the backscattered pressure can then be determined (Eq. 35.2), see text for details. The top panel shows the effect of acoustic pulse amplitude. (A) The peak-positive amplitude of a typical two-cycle imaging pulse is increased from 100 to 500 kPa. (B) The MB radial oscillations become nonlinear when driven at higher amplitudes. (C) The echo response pressure likewise increases nonlinearly owing to faster MB wall velocity. The bottom panel shows the “pulse inversion” contrast imaging mode. (D) A 300 kPa amplitude two-pulse train with an inverted second pulse is delivered to the MB by an ultrasound transducer element. (E) The MB radial oscillations are nonlinear with respect to phase of the driving pulse. (F) The backscattered echoes are also nonlinear with respect to phase and sum coherently to produce a signal (shown is the square root of the sum squared), whereas the linear responses of blood and tissue cancel and no signal would be detected. Simulations were done with Lars Hoff’s BubbleSim code (<https://home.usn.no/lah/Bubblesim/Bubblesim.html>) [12]; R-P with radiation damping; isothermal gas model; Hanning pulse envelope; 2.5 cycle pulse length; 4.0 MHz center frequency; 2.0 μm bubble radius; no shell (all shell parameters set to zero).

used in ultrasound imaging, and resonance can be exploited to generate a very strong echo. Even in the regime of linear oscillations, a resonant MB scatters ultrasound over 10 orders of magnitude greater than a liquid or solid particle of equivalent size [12,15]. The echo of a MB is nonlinear and unique, and it can be readily differentiated from the more linear response of blood and tissue.

The echogenic behavior of a MB for contrast-enhanced ultrasound imaging is demonstrated here by a model simulation based on the Rayleigh-Plesset equation [12]. Fig. 35.2D–F shows results for an uncoated MB undergoing a train of two sequential pulses, where the second pulse is an inversion of the first. The radial oscillations are nonlinear with respect to phase. Thus, a signal is detected from the summed pressure responses. Blood and tissue, on the other hand, scatter linearly, and the pressure echoes from the original and inverted pulses cancel each other. This technique to delineate MBs from tissue is termed “pulse-inversion” imaging. The amplitude can also be modulated to improve further upon the contrast derived from the nonlinear response of MBs in comparison to blood and tissue. Additionally, one may take advantage of harmonic and subharmonic emissions by the MB to improve contrast.

Due to the high backscatter compared to blood and tissue, a clinical ultrasound system is capable of detecting the signature from a single MB—a volume on the order of a femtoliter. This provides an opportunity to detect a single MB adhered to a vascular target within the space of a single voxel, and to distinguish the MB from the surrounding tissue.

Microbubble Stability

An uncoated MB is highly echogenic, but it is unstable owing to surface tension of the gas/liquid interface. In a suspension of MBs, surface tension leads to two destabilizing effects: coalescence and dissolution [15]. For coalescence, surface tension causes an attractive hydrophobic surface force between two uncoated MBs [16]. The hydrophobic force acts in addition to the attractive van der Waals surface force, and the strong attraction drives the two bubbles to coalesce into a single bubble. Coalescence proceeds as the smaller MBs merge to form large bubbles that eventually rise under buoyancy and then pop at the surface. This process is often called “creaming” in the emulsion industry. A surfactant may be added to induce a repulsive surface force (electrostatic double layer, steric overlap, etc.) [15], but it is insufficient to stabilize the MB unless it also diminishes surface tension.

Surface tension (σ) at the curved gas/water interface leads to a net overpressure inside the MB (Δp) given by the Laplace equation:

$$\Delta p = p_b - p_L = \frac{2\sigma}{R} \quad (35.3)$$

where p_b is the pressure on the gas side of the bubble wall. For an uncoated MB, the Laplace pressure (Δp) is on the order of an atmosphere. Henry's Law predicts that this overpressure increases the local solubility of the gas at the MB surface, thus creating a concentration gradient to drive gas diffusion from the core into the surrounding medium. For a suspension of microbubbles, gas diffusion occurs from smaller to larger microbubbles in a process called "Ostwald ripening."

Epstein and Plesset derived an ordinary differential equation for microbubble dissolution [17]. The transport model was formulated by taking a mass balance over the MB and coupling it to the diffusion equation to arrive at the following expression for the MB radius as a function of time:

$$-\dot{R} = \frac{HD}{R} \left(\frac{1 - f + \frac{2\sigma}{p_L R}}{1 + \frac{4\sigma}{3p_L R}} \right) \quad (35.4)$$

where H is the solubility (v/v) of gas in water (Ostwald coefficient), D is the gas diffusivity in water, and f is the ratio of the gas concentration in the bulk medium versus that at saturation. The Epstein-Plesset equation predicts that a free air MB will completely dissolve within a second in saturated water (Fig. 35.3). One approach to increase stability has been to use hydrophobic gases, such as perfluorocarbons, which have water permeation resistances ($H^{-1}D^{-1}$)

that are several orders of magnitude higher than air. The water permeation resistance of perfluorobutane (C_4F_{10}), for example, is over 100-fold greater than that of nitrogen. The mass transport properties of relevant gasses in water are summarized in Table 35.2. Although the use of C_3F_8 or C_4F_{10} can increase MB lifetime by an order of magnitude or more (Fig. 35.3), the surface tension effect drives complete MB dissolution within a minute, which is far too short for a molecular imaging study.

Encapsulation is therefore required to eliminate surface tension and stabilize the MB. Soluble surfactants, such as detergents, may lower the surface tension of a MB, but the large curvature nonetheless results in a large Laplace pressure and complete dissolution within minutes. For appropriate stability on the shelf and in vivo, the MB shell must eliminate surface tension. This can be achieved with phospholipids below their main phase transition temperature. Lipids adsorb to the gas/water interface and become pinned there due to their hydrophobicity. Partial MB dissolution and film compression lead to lipid packing and monolayer formation [18], where the lipids achieve a state of zero net interfacial tension through a balance of attractive and repulsive intermolecular forces [19]. Removing the overpressure eliminates the driving force for dissolution in saturated media [20]. This allows long-term storage of MBs, which are stable for months in a sealed vial.

In addition to eliminating surface tension, the shell may contribute a resistance to gas transport, as modeled by Borden and Longo [21]. Gas permeation through the shell and diffusion in the surrounding medium are modeled as resistances in series, analogous to electrical circuits. The shell resistance is a function of the permeating gas species and the shell composition. In convective flow, where the diffusive boundary layer becomes thin, the shell resistance becomes dominant.

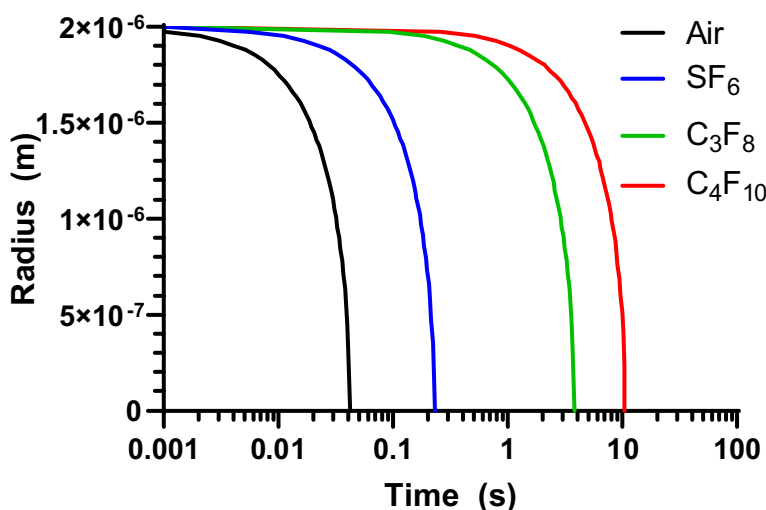


FIGURE 35.3 Uncoated MBs dissolve due to Laplace pressure according to the Epstein-Plesset Eq. (35.4). The MB lifetime can be extended using hydrophobic gases, such as perfluorocarbons. However, an encapsulation that diminishes surface tension is necessary to stabilize the MB long enough for a molecular imaging study. See Table 35.2 for transport properties of the gases presented here.

TABLE 35.2 Gas parameters used in the modeling of MB dissolution time.

Gas	Solubility, H (mL/mL)	Diffusivity, D_w (cm ² /s)
Air	1.7×10^{-2}	2.0×10^{-5}
SF ₆	5.4×10^{-3}	1.2×10^{-5}
C ₃ F ₈	5.2×10^{-4}	0.8×10^{-5}
C ₄ F ₁₀	2.0×10^{-4}	0.7×10^{-5}

The Lipid Shell

The most common MB shell component for USMI has been phospholipid. Several lipid-based ultrasound contrast agents are commercially available worldwide (Table 35.1). Lipid-stabilized MBs are easy to manufacture, biocompatible, and echogenic. A broad library of different lipids is available to provide stability and functionality. Most formulations for targeted MBs consist of three components: a matrix lipid, an emulsifying lipid, and a targeted lipid. The matrix lipid stabilizes the shell by removing surface tension and imparting cohesion. Longer phospholipid acyl chains provide enhanced cohesion through additional van der Waals intermolecular forces [19]. The emulsifying lipid usually contains methyl-terminated polyethylene glycol (mPEG) to inhibit coalescence and passivate the surface [22]. The brush is often formed by mPEGs of 2000–5000 Da molecular weight; shorter mPEGs (e.g., 1000 Da) are unable to stabilize the MB. On the targeting lipid, a polymer spacer is necessary to extend the ligand past the mPEG brush [23].

The lipid shell imparts viscoelastic properties to the microbubble that impact its stability [24] and echogenicity [25]. Models for the lipid shell mechanics have evolved from those used for bulk, polymerlike materials to those specific for surfactant films with infinitesimal thickness by use of surface viscous and elastic terms [26]. These surface terms have been modified to include buckling and rupture effects that capture interesting phenomena, such as “compression-only” behavior [27]. Many experimental investigations have been done with high-speed cameras [13,25], precision acoustics [28], laser-ultrasonics [29], and monodisperse microbubbles [30] to explore and validate these theoretical models and determine their physical constants for lipid shells.

Microbubble Synthesis

The classical method to produce microbubbles quickly, easily, and in high yield is sonication (Fig. 35.4). Sonication involves the vibration of an ultrasonic horn tip at the gas/water interface, where microbubbles are generated and

subsequently coated by liposome adsorption [18]. An alternative mechanical agitation method that can be done at the patient’s bedside is vigorous shaking of a serum vial of lipid solution with a gas headspace, typically with a device like a dental amalgamator. Both sonication and shaking produce polydisperse microbubble sizes, but specific sizes can be easily isolated by a centrifuge [31] (Fig. 35.4C). Additionally, microfluidic flow-focusing devices have been developed to generate monodisperse microbubbles [32,33], although production rates with microfluidic techniques still lag behind sonication and agitation methods.

Ligand Conjugation

MB stability is enhanced by the incorporation of a brush layer of PEGylated lipids, with concentrations of 5–10 M percent of total lipid typically reported. Ligands are incorporated on the distal end of a polymer tether. Architecture refers to the internal structure of the brush layer on the MB shell. Architecture therefore depends on both composition and surface microstructure. Bimodal brushes can be used. A longer spacer arm than the surrounding brush can be used to increase the ligand availability, and therefore adhesion strength [23]. Alternatively, a shorter spacer arm can be used to decrease ligand availability, and therefore immunogenicity [34]. The architecture of the targeting ligand depends on details of the ligand molecule and the linking chemistry. Small ligands (order 100 Da) attached to the distal end of the polyethylene glycol (PEG) chain exhibit dynamics governed by the thermal motion of the polymer spacer. Larger ligands (order 1000 Da), however, significantly change polymer dynamics.

Small molecule ligands can be directly attached to lipids and the ligand-lipid conjugate is then purified prior to their incorporation in MBs. The methods for lipo-PEG-peptide synthesis, such as those employing fluorenylmethoxycarbonyl (Fmoc) chemistry protocols, have been proposed for many years. However, only ~1% of lipid is typically incorporated into the shell during MB synthesis, so this method is not only inefficient, but also presents a competitive binding risk between free lipid particles (liposomes and micelles) and the echogenic MBs. The latter issue can be resolved by washing the MBs with centrifugation, as discussed below.

Large proteins (including antibodies) are often attached through covalent or noncovalent chemistry after the contrast agent is formed due to the harsh conditions that accompany MB creation. The relatively large diameter of a MB necessitates the use of a large number of ligands (100,000 per MB) to ensure adequate coverage. Biotin-streptavidin approaches, in which biotin is attached to the lipid or PEG molecule, have been used commonly due to their simplicity (Fig. 35.5A). A common ligand linkage

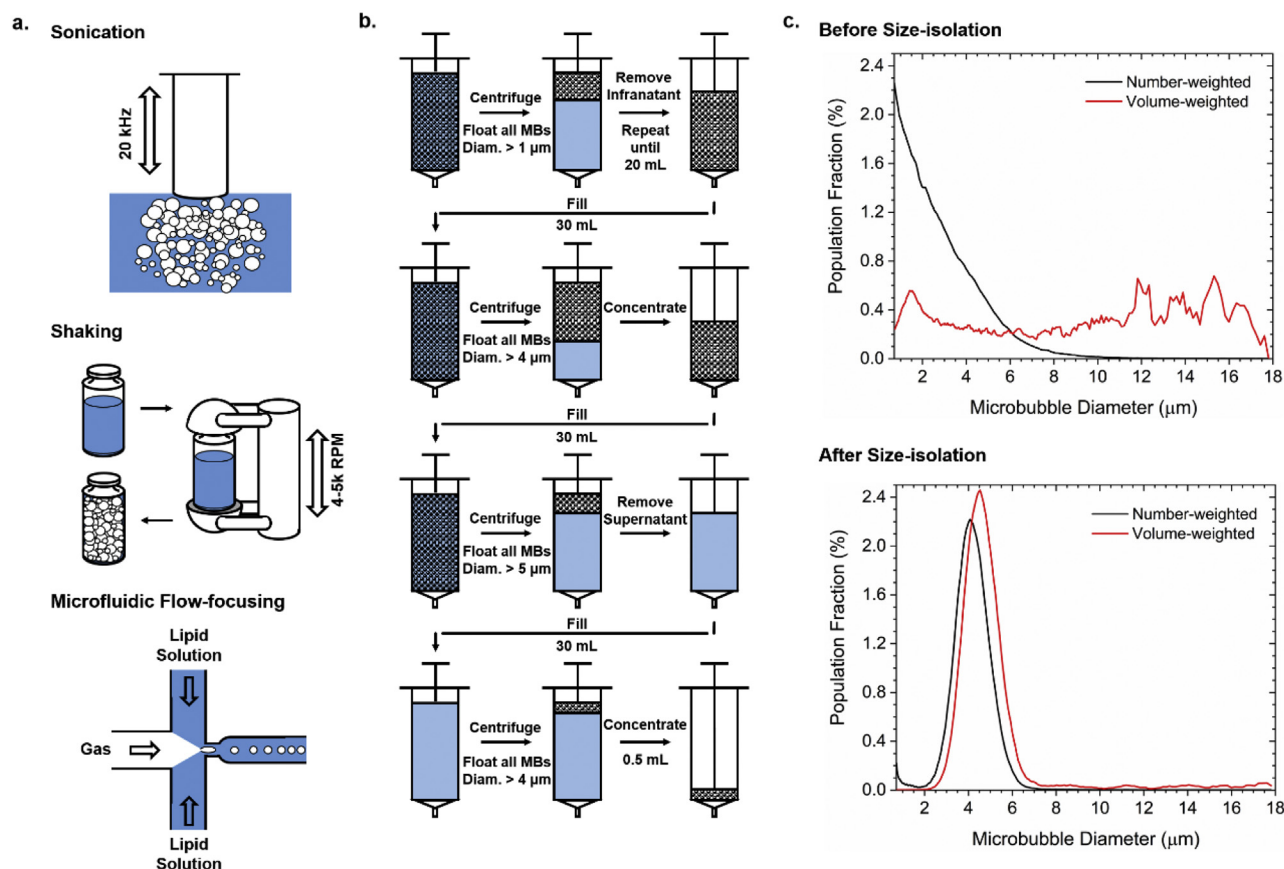


FIGURE 35.4 Microbubble (MB) synthesis and centrifugal sorting. (A) Methods of generating microbubbles. Sonication involves the vibration of a horn tip at the gas/water surface to entrain gas and form microbubbles. Shaking involves the vigorous agitation of a serum vial containing the gas and liquid with surfactant (e.g., lipid) along the long axis of the vial to entrain gas and form microbubbles. Microfluidic flow-focusing involves a gas thread sheathed by liquid with surfactant expanding from constriction and experiencing a capillary instability that leads to regular microbubble pinch-off. (B) A monodisperse microbubble suspension of desired size can be isolated by centrifugation from a polydisperse suspension, such as that formed by sonication. The steps are shown for isolation of a 4–5 μm diameter population. (C) Size distribution measurements of sonication-produced microbubbles before and after centrifugal size isolation following the steps shown in panel (B). Note that successful isolation of a monodisperse suspension requires overlap of the distributions weighted by microbubble number and volume.

motif is biotin-avidin-biotin. Avidin contains four binding pockets and can bind to two or more underlying biotinylated lipids. The avidin molecule has a mass of ~60 kDa and is several times larger than the underlying PEG. Furthermore, antibody molecules often have a mass of ~120 kDa. Thus, the architecture can be viewed as a scaffolding structured with a thin polymer cushion and bulky protein outer layer. In order to use avidin-biotin linkage, an excess of streptavidin is incubated with the MBs, followed by centrifugation to remove excess streptavidin. The biotinylated ligand is then added, the suspension is incubated on a rotator, and the MBs are washed in a final step. The disadvantage of this approach is the immunogenicity of the resulting surface, rendering this approach to be inappropriate for translational studies.

Covalent attachment strategies have also been implemented. The advantages and disadvantages of carboxylic acid-amine approaches, in which a carboxylated lipid

derivative is incorporated into the MB shell and reacts with an amine on the ligand, were described in detail [35]. Potential side products of this reaction include the formation of N-acylisourea, which binds to the MB shell. A more efficient strategy for post-labeling with peptides and antibodies has been developed using a maleimide-thiol approach, in which a maleimide-PEG-lipid is incorporated in the monolayer shell and reacts with a thiol-containing targeting ligand (Fig. 35.5B). Maleimide-reactive groups are more stable in aqueous buffers than an NHS buffer, side reactions are reduced, and a greater percentage of the ligand binds to the target site. However, unreacted maleimide may react with cysteine residues on serum proteins, thus facilitating MB opsonization. For this reason, an extra maleimide capping step with free cysteine is necessary. More recently, a click chemistry approach employing strain-promoted [3 + 2] azide-alkyne cycloaddition (SPAAC) has been used to attach targeting ligands to MBs [36]

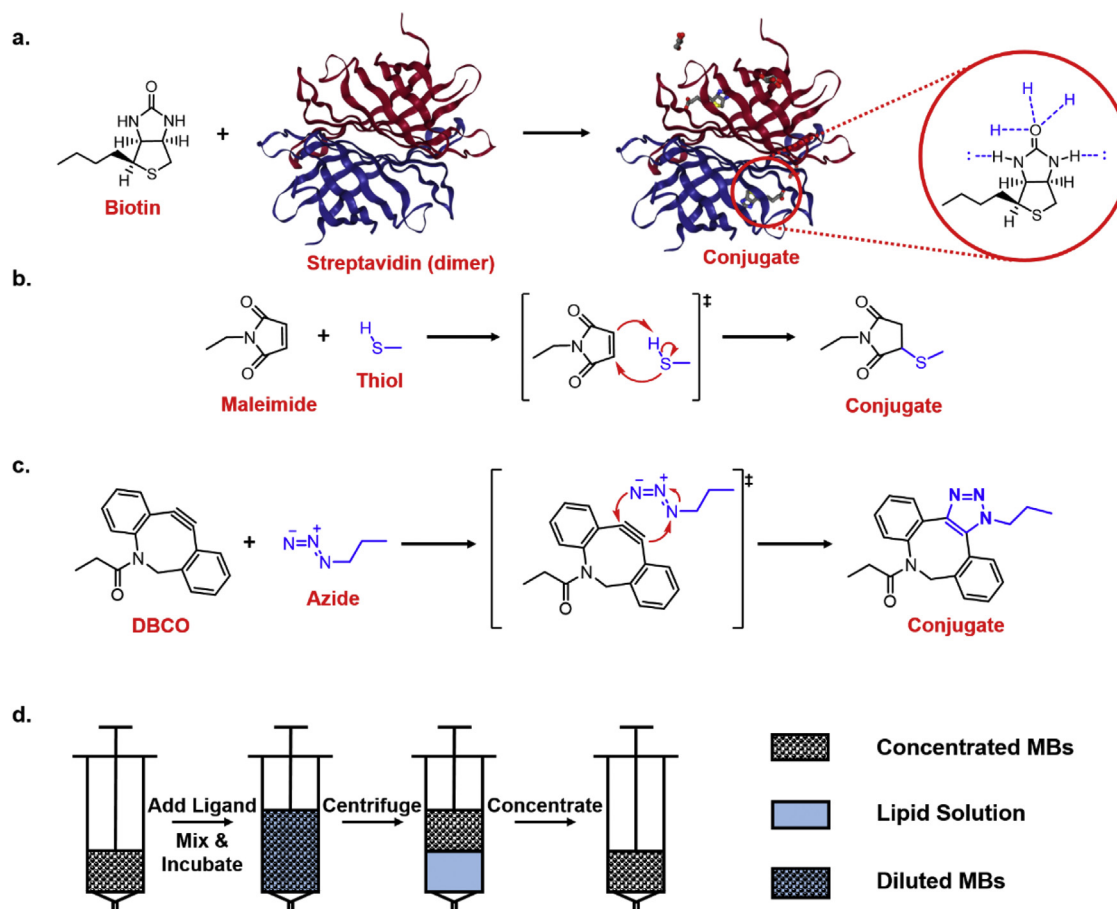


FIGURE 35.5 Ligand conjugation chemistries onto MBs. (A) Biotin-avidin conjugation involves the noncovalent coupling of biotin to one of four binding pockets in an avidin homologue (e.g., streptavidin). Shown are the hydrogen bonds used to capture the biotin molecule. Biotin-avidin coupling is simple to employ, but it leads to an immunogenic surface with limited ligand mobility. (B) Maleimide-thiol conjugation produces a covalent bond that preserves ligand mobility, but unreacted maleimide may react with proteins in the host serum. The arrows denote electron transfer. (C) Click chemistry approach employing strain-promoted [3 + 2] azide-alkyne cycloaddition (SPAAC) also produces covalent bond that preserves ligand mobility, but it avoids the need to cap unreacted groups and is more efficient for bioconjugation [36]. (D) Process flow diagram showing the mixing, centrifugation, and washing steps for ligand conjugation to microbubbles.

(Fig. 35.5C and D). This bio-orthogonal chemistry avoids the need to cap unreacted groups and is more efficient for bioconjugation.

Microbubble Pharmacokinetics

Little is currently known about the proper dose and pharmacology of targeted MBs for use in molecular imaging. Such data has been published for blood pool contrast agents approved for echocardiography (Table 35.1). However, targeted MBs have inherently different and unique surface chemistries owing to the targeting ligands and therefore could exhibit different pharmacokinetics. Clearly, immunogenicity of the targeted MBs is a key concern.

In general, MBs exhibit similar mechanical properties to erythrocytes as they circulate in the blood. Intravital microscopy studies have confirmed that MBs have similar velocities to red blood cells in arterioles, venules, and

capillaries [37]. When they become lodged in a blood vessel, they simply dissolve, deform, and become dislodged without significant vascular effects, as is observed for normal leukocyte plugging.

PET imaging of a lipid-shelled MB demonstrated that accumulation in the lung, liver, and spleen is significant, with some variation according to the model system [38,39]. These organs have capillary beds with large populations of phagocytic cells, such as Kupffer cells in the liver that can remove the MBs from circulation. Circulating white blood cells, such as neutrophils, can also bind to MBs and facilitate their removal. Despite the biocompatibility of the materials used to fabricate MBs, they tend to be decorated with immunological markers present in the blood (e.g., complement proteins or antibodies), which facilitates their detection by the mononuclear phagocyte system. For example, the targeting ligand may have exposed nucleophilic groups, such as hydroxyls and amines that can bind

to the unstable thioester bond on the complement protein C3b. Binding of C3b to the surface of the MB not only changes the ligand binding properties but also marks the MB for immune clearance and proceeds with activation of the complement system and the inflammatory response (see SAFETY section below). Complement activation is of course undesired for molecular imaging, which is primarily used for diagnostic purposes of the unaltered physiology. Thus, it may be advantageous to shield the ligand with an mPEG overbrush, and the ligand can be revealed by ultrasound-induced radial oscillations and acoustic radiation force [34], as discussed later. However, it is also important to consider that MB removal may be enhanced by an adaptive immune response, which results from antibody generation against mPEG itself incorporated into the MB lipid layer. In one preclinical study observing repeated dosing over 28 days, a reduction in nontargeted MB half-life by fourfold was observed in the kidney and surrounding tissue, although MB signal persisted in the liver [40].

Detecting Targeted Microbubbles

For USMI to be effective, the signal from bound MBs must be differentiable from that of unbound, free MBs. The classical approach takes advantage of the rapid deterioration of freely circulating MBs compared to bound MBs [1,41–43]. Following an injection of targeted MBs, a dwell time of 5–10 min is taken to allow targeted MBs to bind while freely circulating bubbles are cleared. During this time, the signal of bound MBs increases while that of free MBs decreases. At the end of the dwell time, an image is taken of the region of interest. Next, a high-mechanical-index “burst” pulse is used to destroy all (free + adherent) bubbles in the field of view. Then a post-burst image acquisition is taken to image only free bubbles as they perfuse into the region of interest (Fig. 35.6). Subtraction of the post-burst from the pre-burst image produces an image

of the bound microbubble contrast. While this technique is effective and easy to implement, data collection is limited by long acquisition times and is restricted to a single time point.

Pharmacokinetic modeling offers an alternative to the classical burst-pulse method, and may offer real-time analysis. In one application, a two-compartment pharmacokinetic model is used to distinguish bound from unbound MBs by comparing the time-intensity and time-fluctuation curves, the latter of which is determined by a motion algorithm [44]. Alternatively, one may employ a random-walk algorithm [45]. The random-walk model adds a ramp function, where the slope represents MB accumulation due to binding in the region of interest.

Finally, various filtering techniques can be taken advantage of to differentiate bound and unbound MBs. Harmonic imaging to separate nonlinear MB signal from linear tissue signal combined with interframe low pass filtering through slow-time to separate adhered from nonadhered MBs serves as a method to isolate bound MBs [46–49]. Decorrelation methods have also been proposed that take advantage of differences in temporal echo characteristics of adherent versus nonadhered MBs [50–52].

Nanodrops

Nanodrops (NDs) are phase-change contrast agents which effectively convert from liquid droplets to gas microbubbles on demand. These contrast agents are formulated with a liquid perfluorocarbon core and encapsulating shell, often with identical components to traditional microbubbles. However, owing to the change in volume with the state of the gas core, a ND will have an initial diameter approximately five times smaller than the resultant bubble (i.e., a 200 nm ND will convert to a 1 μ m MB). Thus, NDs offer the advantages of echogenic MBs in a more compact form. The smaller size of NDs allows them to circulate

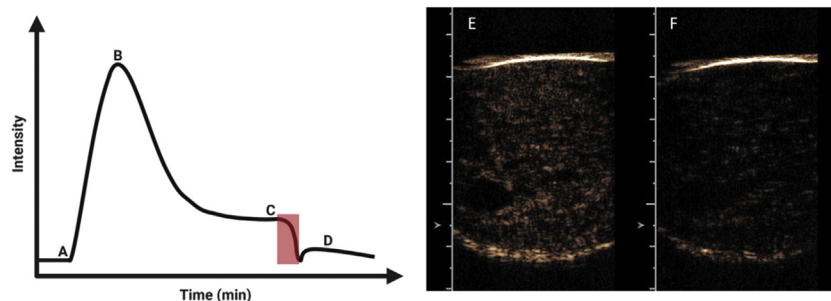


FIGURE 35.6 Representation of image-destroy-image methods for delineating adherent MBs. Time-intensity curve behavior of adhered and nonadhered MBs (A–D). Upon injection (A), MBs flow into the field of view (FOV) until they reach peak enhancement (B), followed by a period of targeted MB accumulation and nonadhered MB clearance (B and C). After sufficient clearance of circulating agents, an image sequence (C) is acquired to measure both adhered and nonadhered MB signal prior to the introduction of a destructive pulse that silences all MBs (red rectangle) in the FOV. After the destructive pulse, any remaining circulating MBs re-enter the FOV and a second image sequence is acquired (D). Calculating the difference in signal intensity between (C and D) provides a measure of bound MBs. (E) Predestruction 2D image of a tumor with bound and freely circulating targeted microbubbles. (F) Postdestruction image of the same tumor where only freely circulating bubbles remain at much lower intensities. Graph (A–D) created with Biorender.com.

longer and to pass through fenestrations in leaky vasculature to access the extravascular compartment of malignant tumors and inflamed tissue. In general, NDs scatter energy according to typical Rayleigh scattering theory. For liquid or solid particles, the incompressibility of the interior phase prevents significant oscillation in the ultrasound field. Thus, the echogenicity of NDs is activated on-demand by acoustic droplet vaporization, whereby the liquid-phase ND is converted to a gas-phase MB. The conversion of NDs to MBs, and subsequent MB destruction or conversion back to NDs, results in ultrasound “blinking” that can be used for advanced imaging techniques, such as super-resolution localization microscopy [53].

NDs can be formed by conventional emulsification techniques, such as homogenization and extrusion, or by condensation of perfluorocarbon gas MBs [54]. Droplet diameters typically range between 100 and 1000 nm, which consequently results in microbubbles on the order of 500 nm to 5 μ m after vaporization. As with MBs, an encapsulation is necessary to diminish surface tension and stabilize the NDs against coalescence and dissolution. Lipid is the most common coating material.

The utility of NDs for molecular imaging was first demonstrated in vitro for targeting the $\alpha_v\beta_3$ integrin on plated human umbilical vein endothelial cells (HUVEC) [55]. NDs conjugated with folate have been targeted to MDA-MB-231 breast cancer cells in vitro, and the NDs were observed to be internalized by the cells prior to vaporization [56]. Folate-carrying NDs were targeted in vivo to SKOV3 ovarian carcinomas and showed remarkable ultrasound contrast after vaporization by low-intensity focused ultrasound [57]. As with MBs, targeted NDs also offer the potential for image-guided therapeutic effects.

Molecular Targeting of Vascular Structures

Due to their restriction to the blood compartment, MBs can be selectively targeted to biomarkers on the luminal side of

the vascular endothelium. As a result, MBs can be used to monitor inflammation, thrombus, or angiogenesis [58–60]. However, dilution in blood and flow conditions, such as margination, may result in low MB concentrations along the vessel wall, where receptor sites are present. Low sensitivity to small numbers of bound MBs compared to free MBs limits the usefulness of USMI. To overcome this challenge, the use of acoustic radiation force to enhance the number of bubbles available for binding at the receptor sites has been explored [42,61–65]. Acoustic radiation force (ARF) enhanced targeting has been shown to drastically improve in vivo USMI signal intensity up to 78% compared to passive targeting alone (Fig. 35.7) [66]. Tailoring MB size to select larger MBs with a larger scattering cross-section per bubble has also been shown to improve the sensitivity of USMI [67,68].

Inflammation

Inflammation plays an important role in many disease processes, including ischemia-reperfusion injury, transplant rejection, and inflammatory diseases in the gastrointestinal tract, heart, liver, and kidneys [69]. Expression of cell adhesion molecules, such as ICAM-1, E-selectin, P-selectin, and VCAM-1, is upregulated with the onset of inflammation. MBs have been targeted to these molecules for monitoring of such inflammatory processes. Selectin-targeted MBs have been used to assess ischemia-reperfusion injury, and even have the capability to identify previously ischemic tissue (Fig. 35.8) [70–72]. MBs targeted to ICAM-1 were used to assess postoperative myocardial tissue acceptance and rejection, with increased signal intensity indicative of graft rejection [73]. Both single-targeted MBs (P-selectin) and dual-targeted MBs (P-selectin and E-selectin) were shown to successfully monitor response to anti-inflammatory treatment [74,75]. Dual-targeted MBs to P-selectin and VCAM-1 showed better adhesion in vitro over single-target agents at physiologically relevant shear stresses, which could improve the detection of vulnerable plaques for atherosclerosis [76]. A triple-targeted

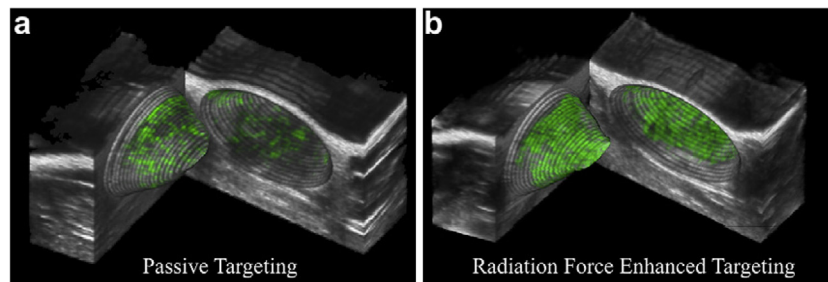


FIGURE 35.7 Demonstration of enhanced MB targeting (green) across a tumor volume with acoustic radiation force (B) compared to passive targeting alone (A). Reprinted from Gessner RC, Streeter JE, Kothadia R, Feingold S, Dayton PA. An in vivo validation of the application of acoustic radiation force to enhance the diagnostic utility of molecular imaging using 3-D ultrasound. *Ultrasound Med Biol.* 2012;38(4):651–660. Copyright 2012, with permission from Elsevier.

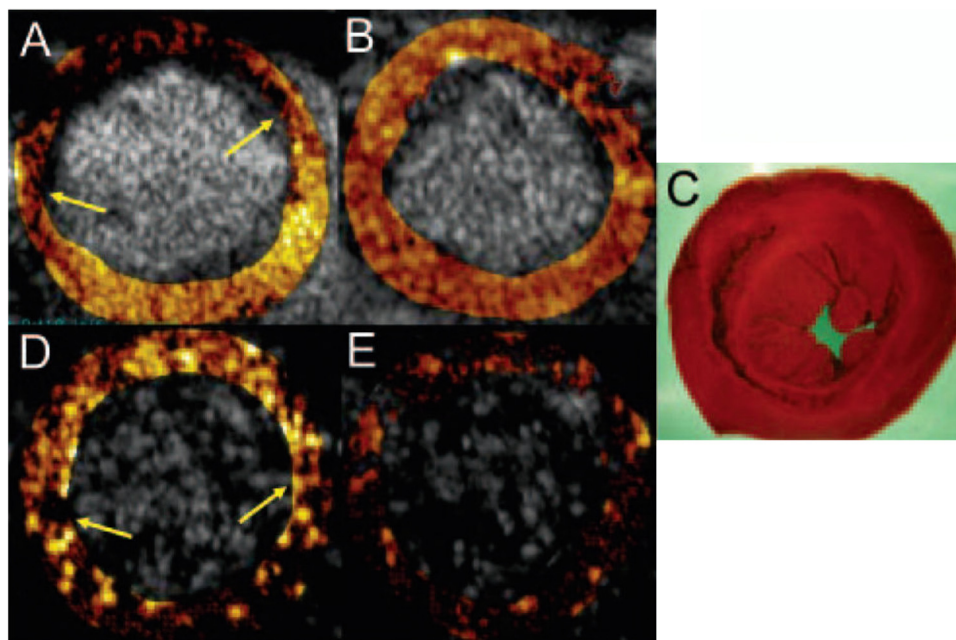


FIGURE 35.8 Myocardial contrast echocardiography during occlusion and after reperfusion with nontargeted (A, B, E) and targeted (D) MBs. During occlusion (A), risk areas for myocardial infarction were identified by lack of perfusion (yellow arrows), which was restored during reperfusion (B). The absence of infarction was confirmed via triphenyltetrazolium chloride staining (C). Thirty minutes after reperfusion there was visible enhancement of previously ischemic areas (yellow arrows) using selectin-targeted MBs (D), which was not observed with nontargeted MBs (E). Reprinted with permission from Wolters Kluwer Health. Villanueva FS, Lu E, Bowry S, Kilic S, Tom E, Wang J, et al. Myocardial ischemic memory imaging with molecular echocardiography. *Circulation*. 2007;115(3):345–352. <https://ahajournals.org/doi/10.1161/circulationaha.106.633917>.

agent for ICAM-1, VCAM-1, and P-selectin showed better in vivo targeting over single- and dual-targeted agents and could be used to identify early-stage atherosclerosis as well as monitor response to therapy [77].

Thrombus

Early detection of thrombus formation, which is typically asymptomatic until reaching advanced stages, can reduce the mortality associated with thrombotic diseases. Targeting activated platelets using MBs has been shown to improve the detection of intravascular and intracardiac thrombus. USMI has been used for visualization of intravascular thrombus and left atrial thrombus and arteriovenous fistulae using a MB with surface ligands targeted to GPIIb/IIIa [78,79]. More recently, a fucoidan polymer MB specific to P-selectin demonstrated in vivo targeting of thrombus [80].

Angiogenesis

New blood vessel formation is a hallmark of cancer, with many cancer types—kidney, liver, breast—overexpressing vascular markers [81]. Much of the focus is on VEGFR2 and $\alpha_v\beta_3$ integrin, two key components of oncogenic signaling pathways. Measuring expression of VEGFR2 and $\alpha_v\beta_3$ integrin on tumor endothelial cells can lead to earlier

cancer detection and intervention [41,82,83]. USMI of angiogenesis to monitor, and even predict, tumor response to treatment has been successfully demonstrated in vivo [84–87]. MBs targeted to $\alpha_v\beta_3$ integrin were used to diagnose tumor responsiveness to antiangiogenic bevacizumab therapy after just 3 days, much earlier than tumor volume or tumor perfusion measurements [76]. Observable decreases in USMI signal of VEGFR2-targeted MBs occurred as early as 1 day following tumor treatment with bevacizumab [86]. MBs targeted to VEGFR2 were also used to indicate tumor response to sunitinib malate, an antiangiogenic agent, after just 1 week of treatment, while changes in volume were not noticeable until after 3 weeks [87]. Furthermore, USMI signal thresholds can be used to differentiate between benign and malignant breast tissue with high sensitivity and specificity [88]. The use of dual-targeted MBs to $\alpha_v\beta_3$ integrin and VEGFR2 have shown greater bound signal intensity over single-targeted MBs and provide better visualization of tumor angiogenesis [43,89].

Safety

Many investigations have demonstrated the safety of MBs, and the U.S. FDA has revised the product label to acknowledge that MBs have an acceptable risk versus benefit ratio [90]. Unfortunately, safety data of MBs may be underappreciated, which likely has led to

underutilization in patient populations likely to derive benefit [90]. MBs are safe in patients with serious cardiopulmonary conditions and intracardiac shunts, and the safety profile of these agents compares favorably with other commonly performed cardiac tests [90]. The risk of severe allergic reactions with MBs is low ($\approx 1:10,000$ doses). However, laboratory personnel should be prepared to recognize and treat these reactions, including emergency administration of intramuscular epinephrine in the most serious cases [90].

The main adverse event for nontargeted MBs is reported to be a type-1 hypersensitivity reaction known as *Complement Activation—Related Pseudoallergy* (CARPA) [91]. In a meta-analysis which included 110,500 patients, the incidence of serious allergic and anaphylactoid reactions immediately after MB administration was estimated at 0.009% and 0.004%, respectively [92]. In addition to MBs, CARPA reactions may also occur with a variety of other drugs and agents including radiocontrast media, nonsteroidal anti-inflammatory drugs, morphine, liposomal agents, and micellar solvents [90]. To date, these rare allergic reactions are the only patient safety risk associated with MB ultrasound contrast agents [90].

The complement proteins are an integral part of the innate immune system, which is the body's main line of defense in detecting and removing foreign pathogens. The immunity pathways converge on the complement protein C3, which is converted to the fragments C3b and C3a by C3-convertase. Normally, these fragments are degraded and recycled by complement control elements to maintain homeostasis. The complement system is activated when C3b binds to the surface of a foreign particle. Immobilized C3b can be recognized by phagocytic cells, and it can interact with the other complement proteins to stimulate humoral immunity and form the membrane attack complex. The C3b fragment contains an unstable thioester bond that binds to an array of nucleophilic groups, such as hydroxyls. This poses a significant challenge for engineering therapeutic devices for injection or transplantation. Surface passivation with mPEG chains has been a major advance in biomaterials science. For MBs, there is also substantial evidence that charged lipids play a role in complement activation. Preferential complement attachment has been shown to lipid-shelled MBs with a net negative charge [93] and those exposing biotin or RGD [34,94]. Complement C3 has also been shown to bind to albumin-encapsulated MBs, mediating adherence of the MB to the vascular endothelium [95]. Complement C3b binding to the targeting ligand can be avoided by use of the buried-ligand surface architecture [34,94,96–98].

Safety of Targeted Microbubbles

Decorating MBs with targeting ligands—such as proteins, peptides or metabolites—could further present chemical groups that bind to C3b and trigger immune activation and a CARPA reaction. The same holds for other colloidal constructs used as imaging contrast agents for other modalities. Clearly, avoidance of an immunogenic response is desirable for targeted contrast agents, not just to minimize hypersensitivity reactions, but also to enable long enough circulation persistence for accumulation at the target. Ideally, the targeting ligand would be hidden from the milieu until the contrast agent reaches the target site, where it is exposed for binding and, through multiple ligand-receptor interactions, results in adhesion. The unique acoustic properties of MBs (radial oscillations and ARF-mediated displacement) have been shown to facilitate a stealth ligand triggered by ultrasound [34,36,94,96–98].

Safety of Insonified Microbubbles

Ultrasound contrast imaging, thus far approved clinically only without a targeting ligand, has been widely shown to be safe and efficacious, with no evidence of cavitation-related biological effects in humans. In vitro, traveling shock waves, fluid shear waves, and liquid jets have been observed, impinging on walls or the cell monolayer, although generally these bioeffects are observed at higher acoustic energies than recommended for contrast-enhanced imaging. Such effects are assumed to be exploitable for MB-induced drug and gene delivery in vivo. These effects increase as a function of frequency, with a dependence on frequency that is greater than that predicted by the mechanical index, and also increase with transmitted pressure. Direct observations of MB oscillation within small blood vessels and models for constrained MB oscillation have been reported, where the expansion of MBs was reported to be diminished within vessels with a diameter on the order of 15 μm or less [99,100].

Clinical Translation

USMI provides a safe, low-cost, and noninvasive method for diagnosing and monitoring disease progression in patients. The advancement of USMI techniques has led to the first human uses with BR55 (Bracco), a clinical-grade MB decorated with a peptide targeted to VEGFR2 [3,4]. A safety and feasibility study of BR55 was conducted in patients with prostate cancer [3]. Patients scheduled at a later date for prostatectomy elected to receive a maximum of two doses of BR55 contrast with ultrasound imaging out

to 10 min post-injection. Patients were monitored for any adverse reactions following contrast administration. USMI was able to visualize 50% of the malignant lesions, and only 17% of patients indicated experiencing nonserious adverse reactions, all of which were resolved without the need for medical attention. This pilot study demonstrated the safety of using BR55 as well as the need for further optimization of contrast dose for prostate cancer imaging.

Additionally, a first-in-human use of BR55 was reported for patients with breast and ovarian lesions [4]. Patients undergoing surgery for either breast or ovarian lesions elected to participate in this exploratory phase clinical trial. A single dose of BR55 was administered to patients followed by USMI image acquisition out to 29 min post-injection. Two radiologists qualitatively assessed focal enhancement, assumed to be the result of bound MBs, in both lesions. Quantitative analysis of signal-to-background ratio was determined for breast lesions. Normal breast tissue could be identified, whereas normal ovarian tissue was difficult to distinguish. Tissue sections were obtained for immuno-histo-chemical (IHC) analysis to assess CD31 and VEGFR2 expression. A comparison of USMI with IHC indicated diagnostic agreement for 86% of the ovarian lesions and 89% of the breast lesions. This study demonstrated the feasibility of USMI to differentiate between benign and malignant lesions and paves the way for further studies investigating the reliability of USMI.

Challenges and Limitations

MBs, which are the most echogenic of ultrasound contrast agents, are large and therefore constrained to the intravascular space. This limits the type of receptor molecules that can be targeted, and restricts molecular imaging to endothelial cell surface phenotypes. However, there are several important vascular markers that are readily accessible to MB targeting, and methods are underway to improve the formulation and use of NDs for extravascular imaging. Challenges for USMI include enhancing target accumulation and improving the delineation of signals from free and bound MBs. More work needs to be done to better understand the in vivo fate of targeted MBs and the possibility of bioeffects induced when these contrast agents are insonified by ultrasound under all conditions, so that imaging can be performed with minimum risk for maximum benefit.

Conclusions and Future Directions

USMI is an emerging field that is enabling real-time, in vivo imaging of several vascular disease processes through longitudinal studies. The BR55 agent from Bracco,

which targets VEGFR2, is paving the way to clinical translation, and two clinical trials in humans have already been reported. Research on the fundamental physicochemistry and acoustic response of MBs has led to revolutionary advances in the rational design of MBs and ultrasound systems that better detect their signature echoes. Commercial systems are currently available, and ultrasound now presents a practical, economical, and effective means of molecular imaging. Several new MB constructs, targeting strategies, comprehensive scanner systems and imaging protocols are currently under development to expand this technology to the clinical arena.

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